

Insight into the Formation of Co@Co₂C Catalysts for Direct Synthesis of Higher Alcohols and Olefins from Syngas

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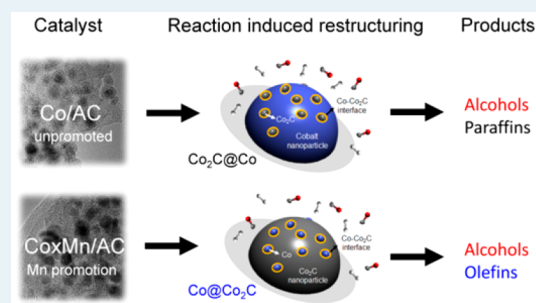
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S Supporting Information

ABSTRACT: Cobalt carbide (Co₂C) has recently been reported to be efficient for the conversion of syngas (CO+H₂) to lower olefins (C₂–C₄) and higher alcohols (C₂₊ alcohols); however, its properties and formation conditions remain ambiguous. On the basis of our previous investigations concerning the formation of Co₂C, the work herein was aimed at defining the mechanism by which the manganese promoter functions in the Co-based catalysts supported on activated carbon (CoMn/AC). Experimental studies validated that Mn facilitates the dissociation and disproportionation of CO on the surface of catalyst and prohibits H₂ adsorption to some extent, creating a relative C-rich and H-lean surface chemical environment. We advocate that the surface conditions result in the transformation from metallic Co to Co₂C phase under realistic reaction conditions to form Co@Co₂C nanoparticles, in which residual small Co⁰ ensembles (<6 nm) distribute on the surface of Co₂C nanoparticles (~20 nm). Compared with the Co/AC catalyst, where the active site is composed of Co₂C phase on the surface of Co⁰ nanoparticles (Co₂C@Co), the Mn-promoted catalysts (Co@Co₂C) displayed much higher olefin selectivity (10% versus 40%), while the selectivity to alcohols over the two catalysts are similar (~20%). The rationale behind the strong structure–performance relationship is twofold. On the one hand, Co–Co₂C interfaces exist universally in the catalysts, where synergistic effects between metallic Co and Co₂C phase occur and are responsible for the formation of alcohols. On the other hand, the relative C-rich and H-lean surface chemical environment created by Mn on the Co@Co₂C catalysts facilitates the formation of olefins.

KEYWORDS: higher alcohols, olefins, Fischer–Tropsch synthesis, Mn promoter, cobalt carbide



INTRODUCTION

Fischer–Tropsch synthesis (FTS) is one of the most versatile process to convert syngas (CO+H₂) derived from coal, natural gas, and biomass into variable chemical feedstock.¹ Apart from the synthesis of high-quality paraffins including gasoline, diesel, and wax, the product slate of FTS can be tuned to synthesize fine chemicals, such as olefins,² C₂₊ alcohols (named as higher alcohols, HAs),³ and aldehydes.⁴ Compared with traditional synthesis strategies, direct synthesis of olefins and HAs from syngas via FTS process is a promising one-pot-one-step route. However, no catalytic system has performed sufficiently well for industrial implementation to date, although great efforts have been made.⁵

Among all the FTS catalysts, Co-based catalysts are always preferred over others due to their better selectivity to long-chain products, lower water gas shift activity, and longer lifetime.⁶ It is well-known that Co⁰ is the active site for the production of paraffins in the FTS process,^{7–9} but recently

cobalt carbide (Co₂C) has been reported to be responsible for the synthesis of C₂–C₄ olefins and oxygenates. Zhong et al. synthesized Co₂C nanoprisms, over which the selectivity to lower olefins was up to 60.8% (C%, CO₂ free) and the ratio of olefins to paraffins was 30. They advocated that the Co₂C nanoprisms exposed a large amount of {101} and {020} crystalline facets, which played a vital role in the formation of olefins.^{2,41} Besides, Co₂C universally exists in the modified Co-based FTS catalysts for the selective synthesis of HAs, such as Co–Cu catalysts, and the selectivity to HAs is relative to the amount of Co₂C according to Volkova et al.'s research.¹⁰ It is acknowledged that the formation of alcohols should be at the interface of two active sites, in which one dissociates CO with the existence of H atoms to produce alkyl chains (C_nH_{2n}*), and

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the other adsorbs CO associatively and facilitates its insertion into the alkyl chain to form alcohols.^{11,12} As a kind of transition-metal carbide, Co₂C has noble-metal-like properties and can adsorb CO molecules associatively.¹³ Thus, the Co–Co₂C interface in the catalysts formed by Co₂C and adjacent Co⁰ should be the active site for alcohol synthesis, based on our previous experimental and theoretical investigations.¹⁴

However, the nature of Co₂C remains ambiguous at present, and further studies are required. It is reported that the amount of Co₂C in Co–Pt/Al₂O₃ and Co/TiO₂ catalysts under realistic FTS reaction conditions can be neglected,¹⁵ while in the activated carbon (AC)-supported Co–Ca/AC and Co–La/AC catalysts, a large amount of Co₂C was always identified after a FTS run.^{16,17} At present, the effect of promoters on the formation of Co₂C have been studied, for example, Lebarbier et al.¹⁸ advocated that the promotion effect of La in the AC-supported Co-based catalysts should be attributed to the small La oxide clusters in the vicinity of Co nanoparticles. They also pointed out that formation of Co₂C was energetically favorable, especially for the Co nanoparticles with smaller sizes because of the formation of surface carbide. However, considering the kinetic aspect, the activation energy (146 kJ/mol) for carbon atoms diffusing into cobalt is about 3 times higher than that into iron, indicating that the carburization of bulk Co nanoparticles should be slow.⁴ In addition, Co₂C has been considered as a metastable phase during FTS reactions, because of its easy tendency to be hydrogenated to metallic Co and CH₄ at a relatively low temperature.^{19,20} Regardless, although there seems to be mounting evidence that the Co₂C phase is reaction-induced, the specific mechanism concerning the formation of bulk Co₂C has not been clarified, especially the influences of oxide promoters.

As an efficient promoter, Mn has been used to improve the activity and selectivity of Co-based catalysts in FTS reactions, and it is a key composition for the synthesis of olefins and oxygenates in the CoMn catalysts synthesized by the coprecipitation process.^{2,4} However, its promotion effects remain ambiguous, and present research has advocated that a minor amount of alkali must be included in the CoMn catalysts for the formation of Co₂C phase.²¹ On the basis of our previous investigations on the formation and catalytic performance of Co₂C, the mechanism by which Co₂C is formed in the manganese promoted Co-based catalysts supported by AC was studied herein. The catalysts, named by CoxMn/AC, were tested and characterized to investigate the promotion effects of Mn, as well as to construct the structure–performance relationship. Microstructure of active sites was determined by a combination of several characterization techniques, including high-resolution transmission electron microscopy (HRTEM), X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), X-ray absorption near-edge structure (XANES), and extended X-ray absorption fine structure (EXAFS).

EXPERIMENTAL SECTION

Catalyst Preparation. The CoxMn/AC catalysts were prepared by incipient wetness impregnation method using AC as the support. The AC raw material (Brilliant Tech Co. Ltd.) was first washed using HNO₃ and deionized water for several times to remove the impurities, and then smashed and sieved to obtain AC support with 180–250 μm diameter after being dried at 120 °C for 24 h. Typically, 1 g of the AC support was added into 1.4 mL mixed aqueous solution of Co(NO₃)₂·6H₂O (Aladdin Company, 99.99% purity) and Mn(CH₃COO)₂·

4H₂O (Aladdin Company, 99.9% purity). The quantities of dissolved Co and Mn precursors were adjusted to achieve various Mn contents from 0 to 3 wt % (on weight basis) with the loading of Co metal fixed at 15 wt % with respect to the sum of Co, Mn and AC support. Afterward, the catalyst precursor was dried overnight at 60 °C in air and then calcined at an elevated temperature up to 350 °C in Ar atmosphere for 16 h, in order to decompose the precursors into metal oxides. Before catalytic reactions, catalysts were in situ reduced in the fixed-bed reactor in H₂ atmosphere at 0.1 MPa and elevated temperature up to 430 °C for 10 h to ensure the fully reduction of cobalt oxide species.

Catalytic Reactions. Catalysts were tested in a fixed-bed stainless steel plug-flow reactor with a 9 mm inner diameter. At the reactor outlet, a condenser was settled to separate the gas and liquid products. About 2 mL of catalyst was mixed well with 2 mL SiO₂ to achieve isothermal plug-flow conditions. The reactor was heated at a fast heating rate of 5 °C/min in H₂ atmosphere at 0.1 MPa from room temperature to 175 °C, and then syngas (H₂/CO = 2) was fed in. Flow rate of syngas was regulated by a mass flow meter to provide varying gas hourly space velocity (GHSV) from 1000 to 4000 h^{−1}. After that, pressure of reaction system was elevated to 3.0 MPa, and the temperature was raised to 220 °C in a decreased heating rate of 0.5 °C/min. Catalytic activity and product selectivities were investigated after 12 h of stabilization on the basis of gas and liquid composition. The exhaust gas was online analyzed by an Agilent 6890N GC apparatus with a Plot Q packed column and a thermal conductivity detector (TCD) with He as the carrier gas. The aqueous and oil products were analyzed by the Agilent 6890N GC with a HP-5 capillary column and a flame ionization detector (FID) with He as the carrier gas. Aqueous product is analyzed by internal standard method using *sec*-butyl alcohol as the internal standard, and the oil product is characterized by the area normalization method. CO conversion (*X*_{CO}) and product selectivity on carbon basis (*S*_{*i*}) was calculated with the following equations.

$$X_{\text{CO}} = \frac{F_{\text{CO,in}} - F_{\text{CO,out}}}{F_{\text{CO,in}}} \times 100\%$$

$$S_i = \frac{N_i \times n_i}{\sum (N_i \times n_i)} \times 100\%$$

F stands for the gas molar flow rate, *N_i* is the molar fraction of product *i*, and *n_i* is the carbon number of product *i*. The carbon balance was calculated and kept higher than 95%.

The catalytic activity for CO dissociation and disproportionation to form surface carbon and CO₂ was accessed by exposing the reduced catalyst samples with varying Mn content ranging from 0 to 3 wt % to CO at 220 °C and 3.0 MPa in the fixed-bed reactor mentioned previously. About 2 mL of catalyst sample was treated in each run, and the reaction rate was calculated by measuring the concentration of CO₂ in the exhaust gas every 25 min utilizing the Agilent 6890N GC apparatus with a Plot Q packed column and a TCD detector with He as the carrier gas. It is noted that the internal transport limitations in these experiments could be neglected by means of Weisz–Prater criterion,²² and the external transport limitations was eliminated by fixing the GHSV at 4000 h^{−1} on the basis of the results of pre-experiments conducted at different bed loadings and GHSVs.

Table 1. Catalytic Performance of Co/AC and Co_xMn/AC Catalysts with Varying Mn Loadings^a

catalysts	CO conversion (%)	selectivity (C%)					STY g/(kgcat·h)
		CH ₄	CO ₂	RH ^b	olefins	alcohols	
Co/AC	47.5	22.9	0.7	47.1	14.7	14.6	92.4
Co0.5Mn/AC	40.5	8.6	1.8	29.0	41.4	19.2	174.3
Co1Mn/AC	29.1	8.1	2.4	29.6	38.5	21.4	119.0
Co2Mn/AC	14.3	9.7	3.1	25.7	41.7	19.8	50.4
Co3Mn/AC	10.5	10.3	2.9	27.6	41.7	17.5	42.4

^aCatalytic tests were conducted at 220 °C, 3.0 MPa, GHSV = 2000 h⁻¹, and a H₂/CO ratio of 2.0. ^bRH stands for C₂₊ paraffins.

Catalyst Characterization. The specific surface area, average pore diameter, and pore volume of activated carbon support and catalysts were measured by N₂ physisorption method using a Quantachrome Autosorb-1 instrument. About 20 mg of sample was loaded into a 9 mm diameter quartz rod and then weighed carefully. Sample analysis was performed in N₂ atmosphere and 77 K, and the system was heated up to 200 °C and evacuated overnight before conducting the measurement.

CO and H₂ pulsed chemisorption measurements were acquired using a Zeton Altramira AMI-300 instrument. About 100 mg of calcined catalyst sample was placed in a U-shape quartz sample tube and then slowly ramped to 430 °C in a flow of 10% H₂/Ar mixture for the reduction of cobalt oxide species. After reduction, catalyst samples were swept by a flow of Ar at 430 °C for 2 h to remove the adsorbed H₂ on the surface of Co nanoparticles. Then the sample tubes were cooled to 50 and 100 °C, at which CO and H₂ uptakes were measured as reported previously,^{23,31} respectively. The total CO uptake was used to calculate the number of exposed Co metal atoms on the surface of Co nanoparticles based on the published methods, in which the CO/Co adsorption stoichiometry was 1/1.²³ The extents of Co reduction in the reduced catalyst samples were accessed by H₂ temperature-programmed reduction (H₂-TPR) using the same instrument as the pulsed chemisorption measurements. About 100 mg of calcined catalyst sample was packed in a U-shape quartz sample tube and then flushed with Ar at 120 °C for 2 h before cooling to 50 °C. The sample was heated to 800 °C at a speed of 10 °C/min in a flow of 10% H₂/Ar mixture. The H₂ consumption was detected by a TCD detector.

The spent catalysts were prepared for HRTEM and EDS characterization by drop-casting ethanol suspension of ground catalysts onto lacey carbon Cu grids as described previously.²⁴ Images of the catalyst samples were acquired with an accelerating voltage of 300 kV using a Tecnai G2 F30 instrument at the State Key Laboratory of Catalysis, Dalian Institute of Chemical Physics, Chinese Academy of Sciences (SKLC-DICP). Particle size distributions for each sample were determined by calculating the mean diameter of approximately 200 nanoparticles.

Powder X-ray diffraction (XRD) patterns were collected using PANalytical X'Pert³ Powder diffractometer with Cu K α radiation in our laboratory. The instrument was operated at 40 kV accelerating voltage and 40 mA current in a scanning angle (2 θ) range of 30–70° at a speed of 2°/min. Catalyst samples were characterized directly after being packed in the glassy sample cell. Diameter of Co nanoparticles was calculated by Sherrer equation. In the equation, k stands for the Sherrer constant, λ is the wavelength of X-ray, B is the full width at half maximum (fwhm) of the diffraction peak, and θ is the angle of diffraction.

$$D = \frac{k\lambda}{B\cos\theta}$$

All of the XRD patterns were analyzed by the X'pert HighScore Plus 2.0 software with the JCPDS library.

In Situ X-ray Adsorption Near-Edge Spectroscopy (XANES) and Extended X-ray Absorption Fine Structure (EXAFS) Spectroscopy. The data of XANES and EXAFS were collected at BL14W1 in Shanghai Synchrotron Radiation Facility (SSRF) in transmission mode at beamline 10-BM. Catalyst samples were first pressed into pellets with 0.1 mm thick together with LiF, and then placed inside a stainless steel in situ cell which was surrounded by a heater and connected to a mass flow meter. First, XANES and EXAFS spectra of the spent Co1Mn/AC catalyst as well as the unpromoted Co/AC catalyst were measured at the Co K-edge (7712 eV) under ambient conditions. Then the spent Co1Mn/AC catalyst was treated by syngas (H₂/CO = 2) in the in situ cell at 220 °C for a fixed time ranging from 10 to 120 min. Reduction of the Co1Mn/AC catalyst sample was also conducted by heating the in situ cell at 4 °C/min in the flowing 10% H₂/Ar mixture up to 400 °C for 120 min, during which the XAFS spectra were measured every 20 min. For comparison, Co₂C sample prepared following the method proposed in the literature was measured by EXAFS and XANES.²⁵

In Situ X-ray Photoelectron Spectroscopy (XPS). In situ XPS studies were performed in a multimethod UHV system with a base pressure of 7.1 × 10⁻⁵ Pa using a Thermofisher Escalab 250Xi instrument at SKLC-DICP. The calcined catalyst samples were pressed in the silver foil and placed in the in situ analysis chamber and reduced in the flowing H₂ at elevated temperature up to 430 °C for 120 min, before and after which the XPS spectra of catalyst samples were measured. Meanwhile, the spent catalyst samples after 48 h time on stream were also analyzed. The X-ray source was operated with 15 kV acceleration voltage and 10.8 mA emission current using a monochromated AlK α radiation. High-resolution scans (fwhm_(Ag 3d5/2) = 0.65 eV, PE = 20 eV) were made for Co 2p, Mn 2p, C 1s, and O 1s. The data was analyzed by the XPS peak software to subtract the Shirley-type background of the spectra and deconvolute the signal envelopes into components (shown as mixed Gaussian–Lorentzian lines using a nonlinear least-squares curve-fitting procedure) as described by Xiang et al.²⁶ The C 1s = 284.6 eV and Si 2p = 103.4 eV were used for charge correction as the reference energies. The relative ratio of atomic surface compositions was calculated using the formula below.

$$\frac{n_i}{n_j} = \frac{I_i}{I_j} \times \frac{\sigma_j}{\sigma_i} \times \frac{E_{k_j}^{0.6}}{E_{k_i}^{0.6}}$$

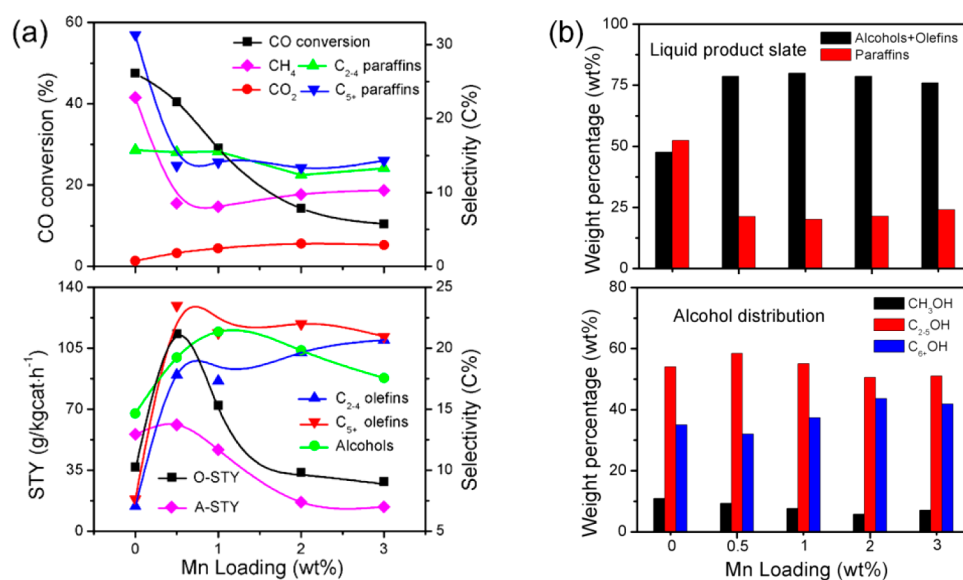


Figure 1. Catalytic performance of the catalysts with varying Mn loadings under the same condition. (a) CO conversion, STY of olefins (O) and alcohols (A) and selectivity to all the products; (b) Liquid product slate and alcohol distribution of Co/AC and CoxMn/AC catalysts. Catalytic tests were performed at 220 °C, 3.0 MPa, GHSV = 2000 h⁻¹, and a H₂/CO ratio of 2.0.

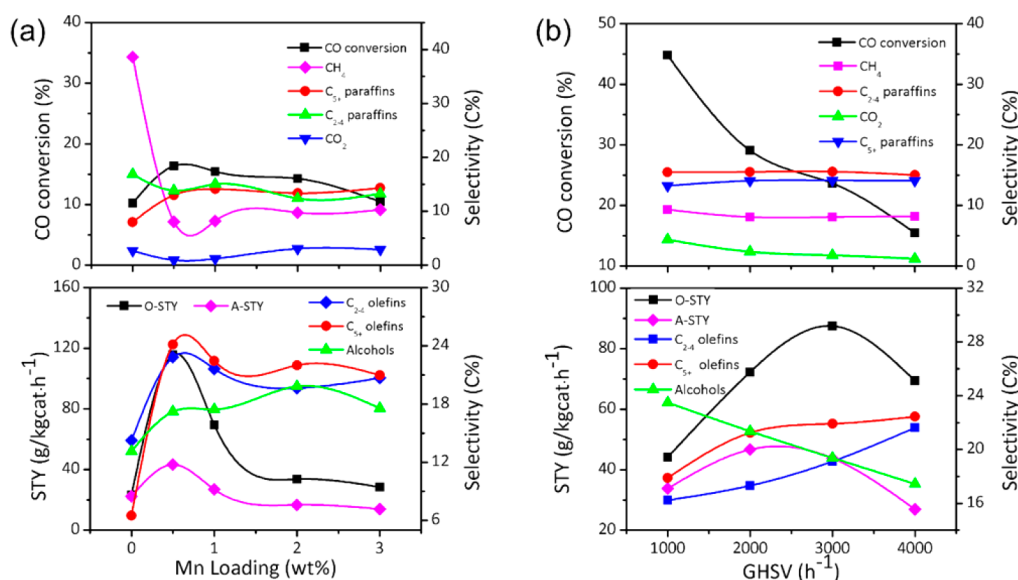


Figure 2. Catalytic performance of the catalysts with varying Mn loadings at similar CO conversion and varying GHSVs. (a) CO conversion, STY of olefins (O) and alcohols (A) and selectivity to all the products of catalysts with varying Mn loadings at similar CO conversions controlled by changing GHSVs; (b) CO conversion, STY of olefins (O) and alcohols (A) and selectivity to all the products over Co1Mn/AC catalyst are shown as a function of GHSVs. Catalytic tests were performed at 220 °C, 3.0 MPa, and a H₂/CO ratio of 2.0.

In the formula, n , I , E_k , and σ stand for the number of atoms, intensity, kinetic energy, and photoionization cross section, respectively. The value of each parameter for Co and Mn can be found in the literature.^{27,28}

RESULTS

Catalyst Test. The catalytic performance of the CoxMn/AC catalysts and the unpromoted Co/AC catalyst were evaluated. The results of catalytic tests are listed in Table 1, in terms of CO conversion, product selectivity, and total space time yield (STY) of olefins and alcohols (O+A). The unpromoted Co/AC catalyst produces a large amount of paraffins (~70%) including a high CH₄ fraction of 22.9%, although the CO conversion is 47.5%, under conditions of 220

°C, 3.0 MPa, GHSV = 2000 h⁻¹, and H₂/CO = 2. Whereas the performance of CoxMn/AC catalysts differs somewhat from that of the Co/AC catalyst in the product slate. First, the O+A selectivity increased greatly to about 60%; second, the selectivity to CH₄ decreased significantly from 22.9% to lower than 10%, accompanying with a slight rise of CO₂ selectivity from 0.7% to ~3%; third, a larger amount of olefins were found in the product (~40%) compared with the Co/AC catalyst (lower than 10%). Intriguingly, a little amount of Mn doping (0.5 wt %) can cause the great change of selectivity, while further Mn addition only results in a decrease of catalytic activity.

However, as it can be seen in Figure 1a, of all the catalysts with and without Mn doping, the selectivities to alcohols are of

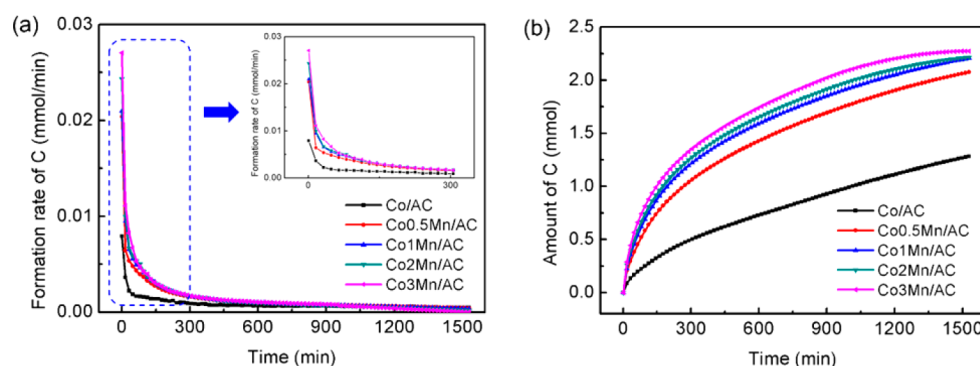


Figure 3. Formation of carbon over Co/AC and CoxMn/AC catalysts. (a) Formation rate of carbon during CO carburization. (b) Total amount of carbon calculated by integrating the formation rate of carbon to time. Experiments were conducted at 220 °C, 3.0 MPa, and GHSV = 4000 h⁻¹.

no significant difference, which varies from 15% to at most 23%. Meanwhile, the mass fraction of HAs in alcohols is higher than 90 wt % for all the catalysts as shown in Figure 1b. It is noticed that Mn doping does not influence the chain-lengthening characteristics of catalysts significantly, since the Anderson-Schulz–Flory (ASF) chain-lengthening characteristics are almost the same, with the calculated α value of ~ 0.65 (Figure S1). As shown in Figure 2a, similar variation tendency of selectivities over CoxMn/AC catalysts with varying amount of Mn can be found when the CO conversions were maintained to the similar value of 10% $\sim$ 15% by varying the GHSV. Nevertheless, the effect of GHSV on the selectivity is shown in Figure 2b based on the Co1Mn/AC catalyst, with respect to its superior compromise between CO conversion and O+A selectivity. It can be seen that the selectivity to alcohols decreased along with the rising of GHSV from 1000 to 4000 h⁻¹, accompanying an increase of olefin selectivity as well as the unchanged selectivity to paraffins. Note that the combined selectivities to alcohols and olefins are $\sim 60\%$ under different GHSVs. According to the results, a trade-off between alcohols and olefins can be identified, which corresponds well with the reported Co/C catalysts.²⁹ In consequence, we suggest that the formation of alcohols/olefins and paraffins may follow different routes, and further investigations should be performed. Moreover, CO conversion and product selectivity as a function of time on stream over Co1Mn/AC catalyst were investigated as shown in Figure S2. After 120 h of time on stream, CO conversion decreased slightly from 29% to 25%, and the catalytic selectivity remained almost unchanged.

Catalytic Activity for CO Dissociation and Disproportionation. Effect of Mn promotion on the formation of carbonaceous species, including Co₂C, carbide carbon, and graphitic carbon,³⁰ was investigated by exposing the reduced Co/AC and CoxMn/AC catalysts to CO atmosphere at 220 °C and 3.0 MPa based on the method reported by Johnson et al.³¹ Formation rate of carbonaceous species was accessed by recording the concentration of CO₂ in the exhaust gas measured by a GC apparatus with a TCD. As shown in Figure 3a, a fast carburization rate was seen in the early stage of reaction, and the rate dropped fast in the first 300 min and remained a low value in the following 1200 min. It can be seen that the addition of Mn lead to the significant increase of the formation rate of C, especially at the beginning of the reaction (0–300 min). The total amount of C formed was calculated by integrating the formation rate to time, as shown in Figure 3b. It is suggested that few amount of Mn (0.5 wt %) results in the

obvious promotional effect while more Mn addition has no further significant influence. The ratio of C to Co after 25 h reaction is listed in Table 2, in which the Co/AC sample had a

Table 2. Ratio of C to Co after 1500 min of Carburization

	Co/ AC	Co0.5Mn/ AC	Co1Mn/ AC	Co2Mn/ AC	Co3Mn/ AC
carbon (mmol)	1.28	2.08	2.21	2.22	2.27
Co (mmol)	3.38	3.37	3.38	3.38	3.37
C/Co ratio	0.38	0.62	0.65	0.66	0.67

C/Co ratio of only 0.38, lower than the stoichiometric ratio of Co₂C (0.50), indicating an insufficient carburization. Whereas for the Co0.5Mn/AC sample, the C/Co ratio was 0.62, showing that most of the metallic Co was carburized, and some excess surface carbonaceous species were formed simultaneously. Moreover, catalysts with more Mn loadings have similar C/Co ratios as that of the Co0.5Mn/AC sample.

Textural Properties. To access the nature of catalysts, the combination of characterization techniques was utilized. First of all, textural properties acquired from N₂ physisorption of the calcined catalyst samples are listed in Table 3. The BET surface area decreases from 991 to 700 m²/g as the Mn loading increases from 0 to 3 wt %, accompanying with the slight decrease of total pore volume. It is revealed that a pore blockage may be caused by the residual carbonaceous species formed by the decomposition of Mn(CH₃COO)₂ precursor in the calcination process.

Chemisorption. The data of H₂ pulsed chemisorption of unpromoted Co/AC and Mn promoted CoxMn/AC catalysts are presented in Table 4. The results demonstrate that the AC-supported Co-based FTS catalyst (Co loading 15 wt %) has much smaller H₂ uptake amount than the SiO₂-supported Co catalysts (Co loading 10 wt %) reported elsewhere,³¹ in which the H₂ uptake amounts are about 0.02–0.05 mmol/g. Otherwise, it is noticed that the addition of Mn causes a further decrease in H₂ uptake over CoxMn/AC catalysts as well as the Mn loading increases. The prohibition effect of Mn promotion on H₂ adsorption over Co nanoparticles has been reported by Johnson et al., in which they advocated that the rationale behind this phenomenon should be the coverage of Co metal surface by MnO_x.³² Whereas in the CoxMn/AC catalysts presented herein, a minor amount of Mn addition (0.5 wt %, the ratio of Co and Mn atoms is about 30) can cause a sharp drop of the H₂ uptake. Meanwhile, further increase of Mn doping (from 0.5 to 3 wt %) results in a monotonous decrease

Table 3. Textural Properties of Co/AC and CoMn/AC Samples Characterized by N₂ Physorption^a

samples	A_{Lang}^b (m ² /g)	A_{BET}^c (m ² /g)	A_{micro}^d (m ² /g)	V_{total}^e (cm ³ /g)	V_{micro}^f (cm ³ /g)	D (nm)	D_{meso}^g (nm)	D_{micro}^h (Å)
AC	1662	1114	633	0.79	0.26	2.9	3.9	5.5
Co/AC	1553	991	482	0.79	0.20	3.2	3.4	5.5
Co0.5Mn/AC	1208	825	475	0.47	0.19	3.1	3.2	5.6
Co1Mn/AC	1254	855	506	0.62	0.21	2.9	3.8	5.6
Co2Mn/AC	1118	773	474	0.56	0.19	2.9	3.1	5.6
Co3Mn/AC	1097	700	333	0.60	0.14	3.4	3.7	5.6

^a A_{Lang} : Langmuir surface area; A_{BET} : BET surface area; A_{micro} : micropore surface area; V_{total} : total pore volume; V_{micro} : micropore volume; D : average pore size; D_{meso} : average pore size of mesopore; D_{micro} : average pore size of micropore. ^bThe surface area was calculated by Langmuir method. ^cThe surface area was calculated by BET method. ^dThe surface area of micropores was calculated by t-plot method. ^eThe total pore volume was obtained from N₂ amount adsorbed at a relative pressure close to unity. ^fThe volume of micropores was calculated by t-plot method. ^gThe average size of mesopores was calculated by BJH method. ^hThe average size of micropores was calculated by HK method.

Table 4. H₂ and CO Chemisorption Characterization Data for Co/AC and CoMn/AC Samples

samples	H ₂ uptake ^a (μmol/g-cat, STP)	CO uptake ^b (μmol/g-cat, STP)	dispersion ^c (%)	Co (0) ^d (%)
Co/AC	34.0	146.5	5.7	84
Co/SiO ₂ ³¹	51 ± 6	—	9 ± 2	79 ± 5
Co0.5Mn/AC	8.3	111.6	4.4	72
Co1Mn/AC	7.2	105.4	4.1	65
Co2Mn/AC	4.3	92.6	3.6	53
Co3Mn/AC	2.2	87.6	3.4	40
Co1Mn/SiO ₂ (Mn/Co molar ratio = 0.1) ³¹	21 ± 5	—	9 ± 3	64 ± 5

^aDetermined by H₂ pulsed chemisorption. ^bDetermined by CO pulsed chemisorption. ^cBased on CO/Metal = 1/1. %Dispersion = $[0.5893 \times (\mu\text{mol/g CO})]/[(\% \text{metal}) \times (\text{reduction percentage})]$. ^dReduction degree, based on H₂ consumption in TPR.

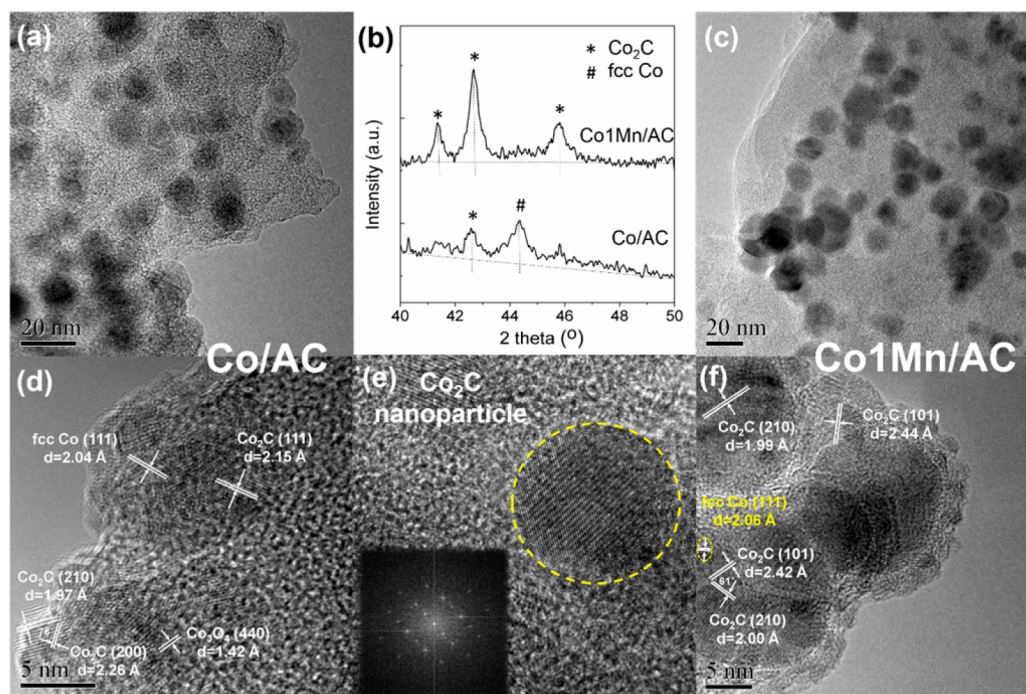


Figure 4. Microscopy of Co/AC and Co1Mn/AC catalysts after reaction. HRTEM images of (a,d) spent Co/AC and (c,f) spent Co1Mn/AC. (b) XRD patterns of spent Co/AC and spent Co1Mn/AC catalysts. (e) HRTEM image of Co₂C nanoparticle in the spent Co1Mn/AC catalyst.

of CO uptake. Accordingly, the dispersion degree of Co decreases with Mn loading amount increasing. Moreover, the prohibition of Mn on the reduction of cobalt oxide species was also found in the catalysts, because the Co reduction degree listed in Table 4 decreases monotonously as more Mn promoter was added. The percentage of Co⁰ for each catalyst

sample was calculated based on the H₂-TPR profiles, whose results are depicted in the Supporting Information (Figure S2).

Microscopy. A series of characterization techniques were employed to identify the microstructure of catalyst active sites before and after reaction. Combined HRTEM images and XRD patterns of spent Co/AC and Co1Mn/AC catalysts after 48 h

time on stream in syngas atmosphere ($H_2/CO = 2$), 220 °C and 3.0 MPa are depicted in Figure 4. According to Figure 4a,d and c,f, the active sites of both Co/AC and Co1Mn/AC catalysts should be cobalt species with spherical morphology, while the microstructures of the active sites in the Mn promoted and unpromoted catalysts are different. In the Co/AC catalyst, Co_3O_4 and face-centered cubic cobalt metal (fcc-Co) seems to coexist with the Co_2C phase, as revealed by the interplanar distances. The interplanar distances of 1.42 and 2.04 Å match well with the (440) and (111) planes of Co_3O_4 and fcc-Co, respectively. And the 2.15 Å of crystalline distance is attributed to the (111) plane of Co_2C . Meanwhile, the interplanar distances of 1.97 and 2.26 Å with an angle of 76° are ascribed to the (210) and (200) plane of Co_2C , respectively. In contrast, a large amount of Co_2C phase is found in the Co1Mn/AC catalyst, while almost no cobalt oxide species or metallic Co phase can be found through HRTEM. Although some Co species with about 2.02 Å of crystalline distance may be assigned to fcc-Co (marked by yellow circles in Figure 4 and Figure S4), but more evidence should be provided to validate its existence.

Bulk structures of Co species in these catalysts were then determined by XRD patterns shown in Figure 4b, in which the major diffraction peaks of 2θ at 41.4, 42.7, and 45.8° correspond to the (020), (111), and (210) planes of Co_2C (JCPDS 05-0704), and the one at 2θ of 44.4° is the (111) plane for fcc-Co metal phase (JCPDS 15-0806). It can be seen that neither Co metal phase nor CoO, but only Co_2C peaks can be found in the spent Co1Mn/AC catalyst. While in the spent Co/AC catalyst, a large fraction of fcc-Co metal is identified to coexist with the Co_2C phase. This result corresponds well with the HRTEM images, implying the significant disparity between the unpromoted Co/AC catalyst and Co1Mn/AC catalysts (i.e., the active sites of Co/AC catalyst should be the mixture of Co metal and Co_2C phase) while the ones of Co1Mn/AC catalysts may be Co_2C nanoparticles as shown in Figure 4e. Moreover, other spent Co1Mn/AC catalysts with varying amounts of Mn in the range from 0.5 to 3 wt % have similar structures as the Co1Mn/AC catalyst, according to the additional HRTEM images (Figure S4) and XRD patterns (Figure S5).

The average particle diameters of all the spent catalysts were measured by HRTEM and calculated on the basis of the results of XRD. As compiled in Table 5, the Co/AC catalyst has the smallest particle size of ~11 nm, while the Mn promoted catalysts have much larger nanoparticles of ~20 nm. The change of particle size may partly be attributed to the expansion of crystalline planes caused by carburization. The result corresponds well with the obvious structural change of the

active site after Mn doping, that is, from a combination of Co and Co_2C (Co/AC) to Co_2C nanoparticles (Co1Mn/AC).

Interestingly, of all the spent Co1Mn/AC catalysts, almost no Mn species can be found, either in HRTEM images or in XRD patterns. It is most probable that Mn species disperse uniformly on the surface of catalysts in the case of low Mn loadings.³¹ The STEM-EDS elemental map (Figure S6) validated that in the spent Co1Mn/AC catalyst, Mn promotion dispersed randomly on the surface of AC support without significant agglomeration.

Reaction-Induced Formation and Microstructure of Co_2C Nanoparticles. To fully understand the nature of Co_2C nanoparticles in the Co1Mn/AC catalysts, we then used in situ extended X-ray adsorption fine structure (EXAFS) and X-ray adsorption near-edge structure (XANES) to determine the fine structure of catalyst samples before and after reactions. The operating temperature of in situ cell of XANES is limited to 400 °C, which we suggest is enough for the fully reduction of cobalt oxide species proved by in situ XRD patterns shown in Figure S7. The normalized Co K-edge XANES spectra of the calcined Co1Mn/AC catalyst sample treated in H_2 atmosphere at 400 °C are shown in Figure 5a, as well as the spectra of Co foil and Co_3O_4 references. The intensity of the peak at 7712 eV, which is attributed to the 1s–3d transition in Co^0 , increased monotonously from 0 to 160 min. It is obvious that the metallic character of catalyst sample was getting greater during H_2 treatment. Moreover, in the Fourier transform curves depicted in Figure 5b, the two main peaks at 1.53 and 2.45 Å correspond to the nearest Co–O and the next-nearest Co–Co coordination in Co1Mn/AC-Calcined sample, respectively. It is seen that the Co–O coordination shell decreased fast during H_2 treatment, accompanying a fast increase of Co–Co coordination shell.

The fitting data of Co K-edge EXAFS are listed in Table 6, from which the Co–Co coordination shell of reduced Co1Mn/AC catalyst (denoted by Co1Mn/AC-Red) is found to be 7.8, while the coordination shell of fcc-Co foil reference is 12. Whereas, Co–O coordination shell remains 1.8 in the Co1Mn/AC-Red, indicating that some cobalt oxide species coexists with metallic Co phase. The result corresponds well with the inspection of Co^0 percentage measured by H_2 -TPR, and the insufficiency of reduction exists universally in the Co-based catalysts. Regardless, in the freshly reduced Co1Mn/AC catalyst, the Co species should be the metallic Co nanoparticles based on the results of in situ XANES and EXAFS.

In order to determine the change of active site structures after reaction, and to inspect the restructuring of active sites after a FTS reaction, we acquired the Co K-edge XANES spectra and EXAFS fitting curves of the spent Co1Mn/AC catalyst sample after 48 h time on stream (denoted by Co1Mn/AC-Spent). As shown in Figure 5c, oxidation state of cobalt in Co1Mn/AC-Spent is initially close to that in the Co_2C sample reference. Meanwhile, the fitting curve and data of Co K-edge EXAFS analysis are shown and listed in Figure 5d and Table 6. All the spectra of the Co1Mn/AC-Spent sample were best fitted using a combination of Co_2C and Co^0 , which contains a large amount of metallic Co phase and differ from the freshly reduced Co1Mn/AC-Red sample. The coordination shell of metallic Co (Co–Co1, 2.50 Å) decreased significantly from 7.8 to 1.6, showing that the particle sizes of cobalt were much smaller after reaction. Accordingly, we conclude that a reconstruction of active site structure in the Co1Mn/AC catalyst occurred during the FTS run, in which a large amount of metallic Co phase converted to Co_2C phase. This was further

Table 5. Average Co/ Co_2C Nanoparticle Diameters of Co/AC and Co1Mn/AC Catalysts Characterized by XRD and HRTEM

catalysts	d (nm) ^a	
	XRD	TEM
Co/AC	11.1	11.4
Co0.5Mn/AC	20.4	19.8
Co1Mn/AC	18.7	18.2
Co2Mn/AC	18.4	17.1
Co3Mn/AC	17.9	16.9

^aAverage particle size of cobalt species.

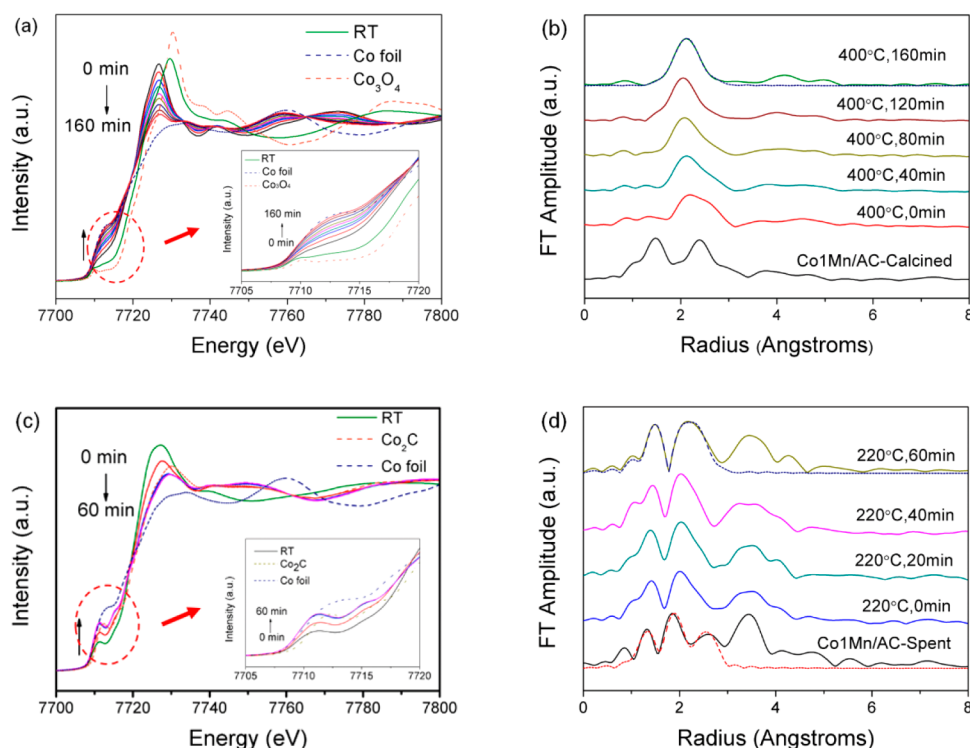


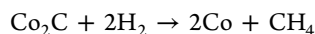
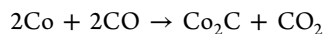
Figure 5. Chemical state characterization of Co1Mn/AC catalyst samples. (a) Cobalt K-edge (7712 eV) XANES spectra of calcined Co1Mn/AC catalyst sample (Co1Mn/AC-Calcined) reduced by H₂ at 400 °C for a maximum of 160 min, and of reference cobalt foil and cobalt oxides; (b) Change of cobalt K-edge EXAFS spectra and data fitting results (dotted line) of the Co1Mn/AC-Calcined sample during H₂ treatment; (c) Cobalt K-edge (7712 eV) XANES spectra of spent Co1Mn/AC catalyst sample (Co1Mn/AC-Spent) treated by syngas (H₂/CO = 2) at 220 °C and 0.6 MPa up to 60 min, and of reference cobalt foil and Co₂C sample; (d) Change of cobalt K-edge EXAFS spectra with the corresponding fitting curve (dotted line) of the Co1Mn/AC-Spent sample during syngas treatment.

proved by in situ XRD, which showed a transition process from metallic Co to Co₂C phase in Co1Mn/AC catalyst under realistic FTS reaction conditions (Figure S8).

Then the Co1Mn/AC-Spent sample was treated by syngas (H₂/CO = 2) at 220 °C for a maximum of 60 min and scanned every 20 min to reveal the structural change under realistic FTS reaction conditions. As shown in Figure 5c, one can see that a fast increase of peak intensity at 7712 eV occurred, showing that the metallic character of the catalyst sample gets greater during syngas treatment. In other words, although Co₂C is the dominated species in the sample, some metallic Co phase still exists at realistic FTS reaction conditions. As shown in the XANES spectra, a fast balance was reached in less than 40 min, and the oxidation state of cobalt in the sample after syngas treatment (denoted by Co1Mn/AC-Syn) is about 1.4 based on the fitting of XANES spectra. Whereas the oxidation state of cobalt in Co foil and Co₂C references are 0 and 1.8, respectively (Figure S9).

The Co K-edge EXAFS curves and fitting data depicted in Figure 5d and Table 6 also indicate that, during syngas treatment, the coordination shell for Co–Co at about 2.50 Å increases from 1.6 to 3.8, along with a decrease in the Co–C/O and Co–Co coordination numbers at about 2.03 and 3.02 Å, attributed to the cobalt oxide (from 4.1/6.6 to 1.7/1.2, respectively). The result suggests that the Co1Mn/AC-Spent catalyst that has some residual metallic Co phase, also underwent transformation under realistic reaction conditions, during which a fraction of Co₂C and CoO are converted into a form with greater metallic character, i.e., cobalt metal phase. On the basis of kinetic analysis, we suggest that the newly formed

Co metal phase in Co1Mn/AC-Spent during syngas treatment should be on the surface of Co₂C nanoparticles, because the trade-off between Co and Co₂C tends to reach a dynamic balance on the surface of Co₂C nanoparticles in the following routes.



After that, the Co1Mn/AC-Syn sample was further treated by syngas for another 60 min at an elevated temperature of 250 °C and 0.6 MPa (Figure S10). Accordingly, the coordination shell of Co–Co at 2.48 Å increased from 3.8 to 4.4 (Table S1). It turns out that the metallic character of Co1Mn/AC catalyst sample got greater at higher temperature, which may be attributed to the conversion of surface Co₂C phase into Co metal ensembles with the aid of H₂. In other words, higher temperature promotes the decomposition of Co₂C phase in syngas atmosphere.

For comparison, the spent Co/AC catalyst sample after reaction (denoted by Co/AC-Spent) was examined by XANES and EXAFS, simultaneously. In this sample, the coordination shell of fcc-Co at 2.48 Å is 6.9, which is much higher than that of Co1Mn/AC-Spent (1.6), even Co1Mn/AC-Syn (3.8), proving that this sample contains larger average crystal sizes of Co metal phase than the Co1Mn/AC catalyst sample. Meanwhile, some Co₂C species still can be found to coexist with metallic Co based on the fitting data, which corresponds to the result of XRD analysis depicted in Figure 4b.

Surface Compositions. In addition to the microstructures, surface compositions of Co/AC and Co_xMn/AC catalysts with

Table 6. Co K-edge EXAFS Fitting Results for Co1Mn/AC Catalyst Samples and the Reference Co Foil, Cobalt Oxide, and Cobalt Carbide^a

sample	shell	N	R (Å)	ΔE_0 (eV)	$\Delta\sigma^2 \cdot 10^3$ (Å ²)	R-factor (%)
Co foil(hcp)	Co–Co	12	2.50	—	—	
Co foil(fcc)	Co–Co	12	2.42	—	—	
CoO	Co–O	6	2.13	—	—	
	Co–Co	12	3.01	—	—	
Co ₃ O ₄	Co–O	6	1.90	—	—	
	Co–Co	6	2.85	—	—	
Co ₂ C	Co–C	1.8	1.90	–5.6	4.00	1.7
	Co–Co1	0.7	2.51	–5.6	6.54	
	Co–Co2	3.4	2.63	–5.6	6.54	
	Co–Co3	2.6	2.79	–5.6	6.54	
	Co–Co4	1.2	3.00	–5.6	6.54	
Co1Mn/AC-Red	Co–O	1.8	2.09	0.9	9.07	0.1
	Co–Co1	7.8	2.43	0.9	11.43	
	Co–Co2	2.4	2.94	3.2	9.07	
Co1Mn/AC-Spent	Co–C/O	4.1	2.03	5.2	11.94	0.7
	Co–Co1	1.6	2.50	5.2	3.06	
	Co–Co2	3.1	2.67	5.2	3.06	
	Co–Co3	2.0	2.83	5.2	3.06	
	Co–Co4	6.6	3.02	5.2	11.94	
Co1Mn/AC-Syn	Co–C	1.7	1.93	–4.3	7.43	0.9
	Co–Co1	3.8	2.48	–6.0	9.78	
	Co–Co2	3.3	2.67	–6.0	9.78	
	Co–Co3	3.0	2.83	–6.0	9.78	
	Co–Co4	1.2	3.01	–6.0	9.78	
Co/AC-Spent	Co–C/O	3.2	2.08	–2.0	11.18	1.1
	Co–Co1	6.9	2.48	–4.6	5.81	
	Co–Co2	1.5	2.67	–4.6	5.81	
	Co–Co3	1.2	2.82	–4.6	5.81	
	Co–Co4	2.9	3.01	–4.6	5.81	

^aN, coordination number; R, interatomic distance; σ , disorder parameter; ΔE_0 , energy shift. All the fitting analysis were performed in the R space, $\Delta R = 1.0$ – 3.0 and $\Delta K = 3.0$ – 12.0 . $S_0^2 = 0.9$ was obtained from fitting the Co-foil standard sample.

varying Mn contents from 1 to 3 wt % were then analyzed by X-ray photoelectron spectroscopy (XPS). The Co 2p and Mn 2p profiles of all the catalysts are compiled in Figure 6a–c and d–f, respectively. As shown in Figure 6a, the Co 2p profiles of the calcined catalyst samples are similar, in which peaks at 779.7, 781.3, and 782.7 eV are located, which represent as Co³⁺ and two kinds of Co²⁺ species in Co₃O₄ and CoO, respectively.³³ Since satellite structure can be found in all the spectra, deconvolution of Co 2p_{1/2} and Co 2p_{3/2} is rather involved. With respect to the Mn 2p profiles shown in Figure 6d, only one peak located at 641.9 eV can be found in the spectra. This peak is assigned to Mn⁴⁺. Obviously there is no significant difference in the XPS profiles among all the calcined catalyst samples.

The calcined catalyst samples were then in situ reduced in H₂ atmosphere at 0.1 MPa for 2 h, and the XPS profiles were collected afterward. Seen in Figure 6b for the Co 2p profiles, a new peak located at 778.2 eV occurred, which is assigned to Co⁰ formed by the reduction of cobalt oxide. It is demonstrated that Co⁰ is the dominating cobalt species in all the samples after reduction, indicating that the metallic Co species is the main composition prior to a FTS reaction, which corresponds to the results of in situ EXAFS and XANES. Simultaneously, the intensity of peak at 779.7 eV was much weaker, and the two peaks at 781.3/782.7 eV were replaced by one at 782.3 eV, which account for the reduction of Co³⁺. Note that all the reduced catalysts have similar spectra except for the Co3Mn/

AC sample, in which a larger amount of residual Co³⁺ and Co²⁺ remained. The prohibition effect of Mn on the reduction of cobalt oxide in CoxMn/AC catalysts was significant only when the amount of Mn loading was high enough. Fitting of Mn 2p spectra, seen in Figure 6e shows two peaks, one is at 642.3 eV and the other is at 640.9 eV, which should be assigned to Mn⁴⁺ and Mn²⁺/Mn³⁺, respectively. It means that a mixture of MnO₂ and Mn₃O₄ existed in the reduced Co1Mn/AC catalyst as proved simultaneously by the linear combination fitting (LCF) of Mn K-edge XANES (Figure S11). The binding energy for the Mn⁴⁺ after reduction differ somewhat from the one after calcination, we suggest that the anomalous positive binding energy shift can be attributed to the interaction between different oxidation states of Mn species.³⁵ Note that the peak intensity of Mn²⁺/Mn³⁺ increases as the amount of Mn was raised. It means that more Mn⁴⁺ can be reduced at higher Mn loadings.

Subsequently, the spent catalysts were measured by XPS and the spectra were compiled in Figure 6c,f. All the CoxMn/AC catalysts have similar Mn 2p spectra, while the Co 2p spectra of Co/AC catalyst differ from that of CoxMn/AC catalysts. As seen in Figure 6c, in the Co/AC catalyst sample, only a large peak at 781.3 eV exists, which is assigned to Co²⁺. Note that neither Co⁰ nor Co³⁺ can be found. We suggest that the residual Co³⁺ transformed into other cobalt species after reaction with the cobalt carbide mediated process, as proved elsewhere.³⁶ On the basis of characterization results obtained

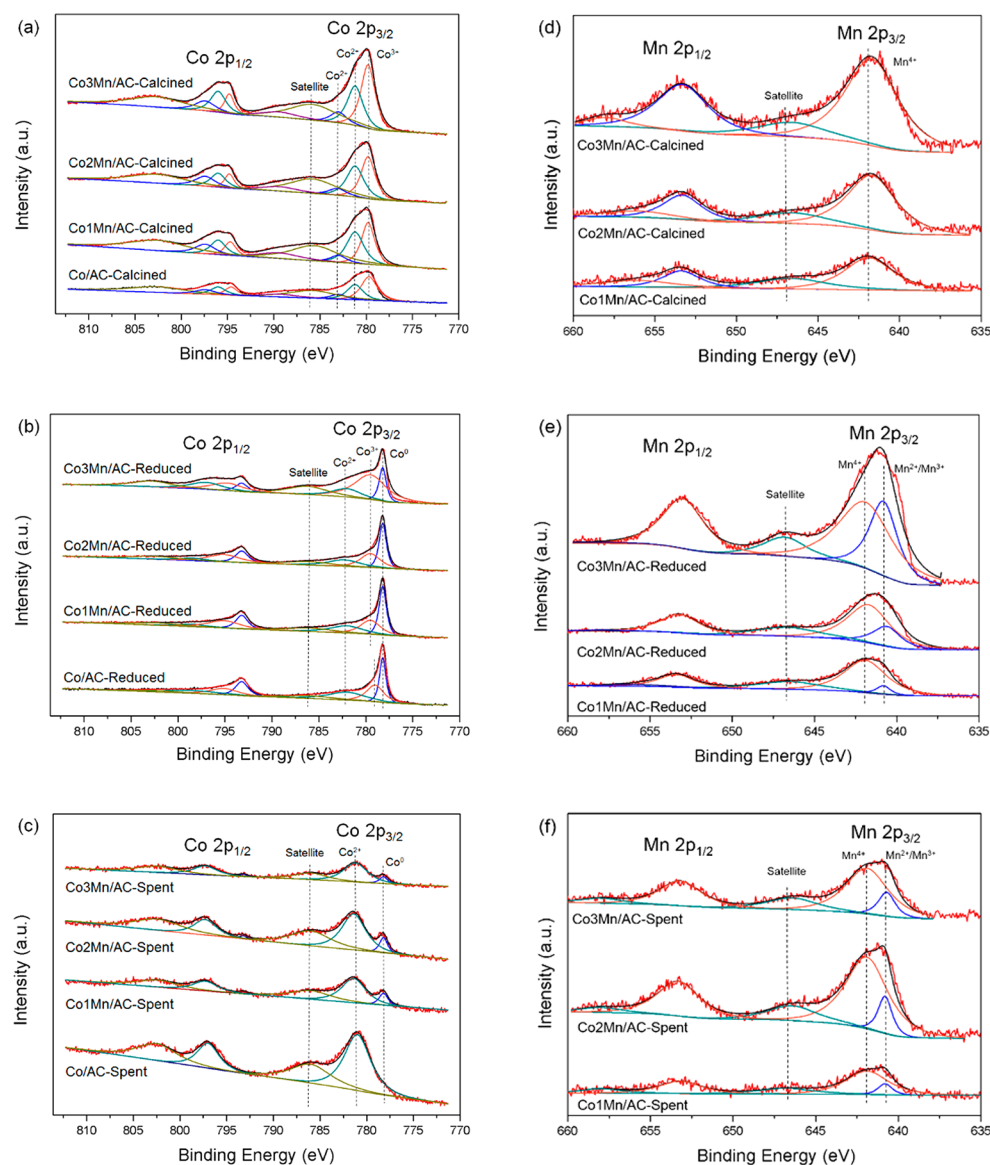


Figure 6. Co 2p and Mn 2p XPS spectra of Co/AC and CoxMn/AC catalysts. (a) and (d) calcined at 350 °C in Ar atmosphere for 16 h; (b) and (e) in situ reduced at 430 °C in H₂ atmosphere for 2 h; (c) and (f) spent after 48 h time on stream at 220 °C, 3.0 MPa, GHSV = 2000 h⁻¹, and a H₂/CO ratio of 2.0.

from HRTEM, XRD and XAFS, one part of Co²⁺ should be attributed to Co₂C for the spent Co/AC sample (seen in the C 1s spectra shown in Figure S12), while the other is the CoO formed by the oxidation of Co⁰ in the sample transferring process (seen in the O 1s spectra shown in Figure S12). It is acknowledged that, the metallic Co phase is easy to be oxidized when exposing to oxygen containing atmosphere, thus it is reasonable to see no Co⁰ in the spent Co/AC catalysts. Whereas, for the spent Co1Mn/AC sample, no obvious peak attributed to Co or Mn oxides could be found according to the O 1s spectra displayed in Figure S12, showing that the Co²⁺ species (the peak at 781.5 eV) found in the Co 2p spectra should be mainly attributed to Co₂C. However, besides the Co²⁺ species similar to that in the Co/AC sample, the spent CoxMn/AC catalysts have a higher binding energy shoulder at 778.2 eV, which is assigned to the Co⁰ phase on the surface of catalyst. The relative contents of surface cobalt species of spent CoxMn/AC catalysts involving Co⁰ and Co²⁺ are compiled in Table S2. Different from the Co⁰ phase, which is easy to be

oxidized in the Co/AC catalyst, the Co⁰ on the surface of CoxMn/AC catalysts is ultrastable. This result proposed direct evidence that metallic Co exists on the surface of Co₂C nanoparticles, which is consistent with the results of in situ EXAFS and XANES. We suggest that the Co⁰ may be protected by a very thin layer of carbonaceous species that was formed in a FTS run via the CO disproportion promoted by Mn doping.³¹ Undoubtedly, this result calls for a more extensive investigation to reveal the rationale behind it. Intriguingly, the catalytic activity and selectivity of CO hydrogenation over Co1Mn/AC catalyst remained almost unchanged after 12 h of passivation in 1% O₂/Ar atmosphere (Table S3), which may be related to the structure of active sites, especially the ultrastable Co⁰ phase.

The ratio of Co to Mn on the catalyst surface before and after reduction was accessed by calculating the relative amount of Co and Mn, according to the XPS spectra of catalyst samples with varying Mn loadings. Seen from the data listed in Table 7, the ratio of Co to Mn on the surface was almost the same as

Table 7. Relative Surface Atom Ratio of Co to Mn before and after Reduction by in Situ XPS Measurement

	bulk	calcined	reduced	spent
Co/AC	—	—	—	—
Co1Mn/AC	14.0	16.6	6.7	2.2
Co2Mn/AC	7.0	11.7	4.7	1.1
Co3Mn/AC	4.7	9.2	3.1	0.7

that in the bulk of catalyst samples. It is thus clear that the distribution of Co and Mn in the calcined catalyst samples was uniform. Investigations on the spatial distribution of Mn over Co-based catalysts have reported that Co and Mn oxides form a mixture after calcination,³⁴ which may be $\text{Co}_{3-x}\text{Mn}_x\text{O}_4$ phase.

Whereas, the Co/Mn ratio underwent a significant decrease of more than a half after reduction, proving that some Mn atoms migrate from the bulk of $\text{Co}_{3-x}\text{Mn}_x\text{O}_4$ phase formed in calcination to the surface of catalysts. The aggregation of Mn species on the surface of catalysts after reduction in Mn promoted Co-based catalysts has been reported in the previous literature.³⁷ The result shows that although the segregation of Mn around Co nanoparticles was not found here, Mn still tends to be distributed on the surface of catalysts after reduction.

DISCUSSION

Effect of Mn Promotion on the Catalytic Performance.

On the basis of catalytic performance results, it is seen that Mn addition strongly promotes the formation of olefins, the selectivity to which increases from about 10% to higher than 40%. Whereas, the promotion effect of Mn on the formation of HAs is not so significant, because the alcohol selectivity increases from 15% to at most 23%. Besides, a strong prohibition effect of Mn on the formation of methane can be found, accompanying with a slight increase of CO_2 selectivity. Actually, Mn has long been known as a kind of effective promotion in the olefin production over Co-based and iron-based catalysts.³⁸ Its mechanism has been thoroughly investigated by Johnson et al., who advocated that the cleavage of C–O bond was promoted through Lewis acid–base interactions between Mn^{2+} on the edge of MnO islands distributed on the surface of Co nanoparticles and the O atom of CO adsorbates adjacent to the MnO islands.³¹ Nevertheless, although Mn doping plays a vital role in the catalytic performance, only a little amount of Mn (0.5 wt %) can cause a significant change in the product slate, while more Mn addition has almost no further influence except for the decrease of catalytic activity. This result corresponds well with the results of CO dissociation and disproportionation experiments. Moreover, this result is consistent with the viewpoint in

previous researches that a small amount of Mn can cause the full promotional effects, while the amount of Mn (0.5 wt %) in the CoxMn/AC catalysts was much smaller than reported (Mn/Co ratio = 0.1).³¹ We suggest that the weak interaction between AC support and Co nanoparticles may enhance the promotional effects of Mn.

Influence of Mn on the Active-Site Microstructure.

Based on the results of characterization techniques, we advocate that the active sites of Co/AC and Mn promoted CoxMn/AC catalysts both underwent restructuring in FTS reactions, from metallic Co to a combination of Co/ Co_2C and Co_2C nanoparticles, respectively. Further investigations showed that the formed active sites have different microstructures, that is, for the Co/AC catalyst as shown in Figure 7a, the active site is composed of metallic Co and Co_2C phase, namely, $\text{Co}_2\text{C}@Co$, in which Co_2C should be on the surface of Co nanoparticles because of the carburization develops preferentially on the surface. With regard to the CoxMn/AC catalysts, in general, the active sites seem to be Co_2C nanoparticles according to HRTEM images and XRD patterns, while they presented metallic character under realistic reaction conditions proved by in situ EXAFS and XANES. Existence of surface Co^0 phase in the spent CoxMn/AC catalysts was subsequently proved by XPS, which is the direct evidence for the metallic Co on the surface of Co_2C nanoparticles. Accordingly, we note, the microstructure of Co_2C nanoparticles in the CoxMn/AC catalysts can be demonstrated as the $Co@Co_2C$ structure depicted in Figure 7b, in which many small Co^0 ensembles distributed on the surface of Co_2C nanoparticles (~20 nm). The size of the Co^0 ensembles should be smaller than 6 nm, because of the limitation of powder XRD analysis. We suggest that the rationale behind the formation of the $Co@Co_2C$ structure under realistic reaction conditions is twofold: one is the formation and stabilization of Co_2C phase, and the other is the formation of surface metallic Co ensembles. Actually, conversion of Co_2C to metallic Co under realistic FTS conditions has been found in the literature,²⁵ in which FTS activity increased as the amount of cobalt carbide decreased. Thus, the formation of stable Co_2C phase is the key for the construction of $Co@Co_2C$ catalysts.

According to the results presented herein, Mn promotion plays the pivotal role in the formation of Co_2C , because no other promoter exists. It is seen that Mn doping greatly enhanced the CO dissociation and disproportionation over CoxMn/AC catalysts, resulting in a relative C-rich surface chemical environment which may facilitate the formation of Co_2C . The promotion effect of Mn on CO dissociation and disproportionation over Co-based catalysts have been reported

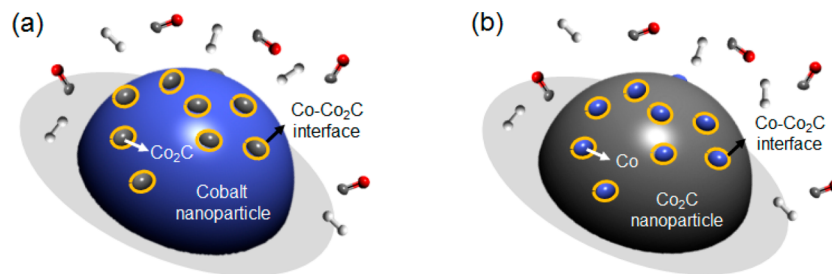


Figure 7. Active-site conceptual frameworks of Co/AC and CoxMn/AC catalysts. (a) Active sites of unpromoted Co/AC catalyst have $\text{Co}_2\text{C}@Co$ structure; (b) Active sites of Mn promoted CoxMn/AC catalysts have small Co metal ensembles dispersed in the bulk and on the surface of Co_2C nanoparticles, noted as $Co@Co_2C$.

by Johnson et al.,³¹ and Zhu et al. advocated that the O vacancy in partially reducible MnO_x may mediate the CO dissociation to form surface carbonate and carbon species.³⁹ Besides, Mn promotion prohibits H_2 uptake on the surface of metallic Co, either by the electronic effect or through covering the surface of Co nanoparticles. Accordingly, we advocate that the abundance of surface carbonaceous species, accompanying the low concentration of adsorbed H atom, result in a relative C-rich and H-lean surface chemical environment on the Co nanoparticles of $\text{Co}_x\text{Mn}/\text{AC}$ catalysts, which promotes the formation of Co_2C and prohibits its decomposition by hydrogenation.

Nevertheless, the formation rate of Co_2C in the $\text{Co}_x\text{Mn}/\text{AC}$ catalysts was very fast, because the induction period for all the catalysts were less than 12 h, according to the stabilization of catalytic activity and selectivity. In kinetic aspects, carburization of Co nanoparticles not only leads to the formation of surface carbonaceous species, but also results in the migration of surface carbon atoms into the core of Co nanoparticles. Previous studies have proved that, although the diffusion energy of carbon atoms into the bulk of Co nanoparticles is high (~ 0.94 eV), it can follow the vacancy-mediated pathway which can decrease the highest diffusion energy barrier to ~ 0.77 eV.⁴⁰ In the $\text{Co}_x\text{Mn}/\text{AC}$ catalysts, vacancy defects of Co nanoparticles have been identified by in situ EXAFS, because the coordination shell of $\text{Co1Mn}/\text{AC}$ catalyst is 7.8, much smaller than that in the Co foil reference ($N = 12$). The “loose” Co nanoparticles would facilitate the migration of carbon atoms, result in the bulk carburization of nanoparticles. On the basis of previous investigations, one can conclude that the formation of the $\text{Co}@ \text{Co}_2\text{C}$ active sites in the $\text{Co}_x\text{Mn}/\text{AC}$ catalysts greatly depends on the specific surface chemical environment (relative C-rich and H-lean) created mainly by Mn doping and the loose structure of Co nanoparticles.

Origin of Structure–Performance Relationship. We note, a minor amount of Mn addition can cause a significant change in both the structure of catalyst active sites and the catalytic performance, and thus, the relationship between the catalyst structure and performance should be established. In fact, it was confirmed by our previous work that the $\text{Co}-\text{Co}_2\text{C}$ interfaces acted as the active site for the alcohol production,¹⁴ which in this work are easily formed in both the $\text{Co}@ \text{Co}_2\text{C}$ and $\text{Co}_2\text{C}@ \text{Co}$ samples independent of their specific structures. Accordingly, the similarity of alcohol selectivity over both the two kinds of catalysts can be explained by the similar densities of $\text{Co}-\text{Co}_2\text{C}$ interface (the ratio of Co_2C to Co is 0.5 and 3.0 characterized by linear combination fitting (LCF) of XANES is listed in Table S4), on which CO molecules dissociatively adsorb on the surface of metallic Co phase to form alkyl chains, and Co_2C acts as the active site for CO associative adsorption, which promotes CO insertion into alkyl chains to form alcohols as depicted in Figure 8.

Whereas, formation mechanism of olefins over $\text{Co}-\text{Co}_2\text{C}$ catalysts remains mysterious, because the active site on which olefins are produced has not been identified. Although Zhong et al. advocated that lower olefins (C_2-C_4) may form on the specific lattice plane of Co_2C ,² no evidence has been provided for the formation mechanism of C_{5+} olefins over $\text{Co}-\text{Co}_2\text{C}$ catalysts. For typical FTS reactions over Co-based catalysts, it is acknowledged that absorbed alkyl chains can either be hydrogenated to form *n*-paraffins, or undergo β -hydrogen elimination to produce α -olefins.⁴² With respect to the effect of Mn on the selectivity of Co-based FTS catalysts, mounting

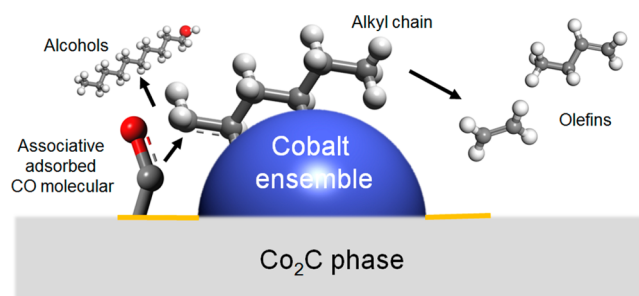


Figure 8. Schematic illustration of alcohol and olefin formation mechanism on the surface of $\text{Co}@ \text{Co}_2\text{C}$ catalyst.

evidence have validated that Mn promotion can increase the selectivity to C_{5+} hydrocarbons by moderating hydrogenation reactions^{34,43} and Dinse et al. advocated that Mn promotion can also increase the selectivity to olefins through suppressing secondary hydrogenation of olefins to paraffins.⁴⁴ Based on the experimental results presented herein (Figure S1), all the products including alcohols, olefins, and paraffins follow an ASF distribution in general, indicating a stepwise polymerization reaction which corresponds well with the typical FTS reaction. Thus, we suggest that the metallic Co phase found on the surface of Co_2C may act as the active site to form alkyl chains ($\text{C}_n\text{H}_{2n}^*$), and the relative C-rich and H-lean surface chemical environment over $\text{Co}@ \text{Co}_2\text{C}$ catalyst plays the pivotal role in the formation of olefins. Nevertheless, compared with the alcohol formation mechanism, which has been acknowledged to be the synergistic effect between Co_2C and metallic Co on their interface, the mechanism for the production of olefins still needs further investigations in future.

CONCLUSIONS

In summary, effects of Mn promoter on the active site structure and catalytic performance of Co/AC catalysts were investigated. Active sites of $\text{Co}_x\text{Mn}/\text{AC}$ catalysts were identified to be the $\text{Co}@ \text{Co}_2\text{C}$ structure, where small Co^0 clusters (< 6 nm) distributed on the surface of Co_2C nanoparticles (~ 20 nm). The structure was formed under realistic reaction conditions through restructuring of Co nanoparticles and facilitated by Mn doping. The function of Mn promoter is to create a relative C-rich and H-lean surface chemical environment that in turn enhances the CO dissociation adsorption and disproportion reaction as well as prohibits the adsorption of H_2 . In addition, the “loose” Co nanoparticles with a large amount of vacancies facilitate the migration of surface carbon atoms into the core of Co nanoparticles, resulting in the fast formation of Co_2C nanoparticles.

Compared with the unpromoted Co/AC catalysts with the active sites as the $\text{Co}_2\text{C}@ \text{Co}$ structure, the $\text{Co}@ \text{Co}_2\text{C}$ catalysts strongly favor the production of olefins, from 10% to about 40%. Whereas, the selectivity to alcohols over the two catalysts were similar (15–23%), and the ratio of HAs to all the alcohols is always higher than 90%. It can be seen that, a minor amount of Mn doping causes a significant change of the product slate, while further Mn addition has little additional influence. This result corresponds well with the structural change of the active site in the $\text{Co}_x\text{Mn}/\text{AC}$ catalysts, suggesting that the active sites structure instead of the Mn promoter is the key factor to determine the catalytic performance.

Considering the differences between the nature of $\text{Co}@ \text{Co}_2\text{C}$ and $\text{Co}_2\text{C}@ \text{Co}$ active sites, the rationale behind their

structures and catalytic performances was twofold. For one thing, formation of alcohols can take place on the interfaces between metallic Co and Co₂C phase, which was abundant in both the two kinds of structures. Thus, the selectivity to alcohols over the two kinds of structures should be similar, which corresponds well with the catalyst test results. For the other, olefin production is supposed to rely on the chemical environment of the catalyst surface, that is, the relative C-rich and H-lean circumstance on Co@Co₂C active sites created by Mn may prevent the hydrogenation of alkyl chains and result in the high olefin selectivity in the FTS process. Whereas, further investigation on the mechanism of olefin formation over Co-based catalysts should be performed in future. Besides, since no Co₂C can be found in the CoMn catalysts synthesized by coprecipitation method without alkali promoters as reported, we suggest that the activated carbon support may also promote the formation of Co₂C, the mechanism of which should be studied in depth as well.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.7b02403.

Co K-edge EXAFS fitting results of Co1Mn/AC-130 sample, surface atom ratio of Co⁰/Co²⁺ of Co/AC and CoxMn/AC catalysts, catalytic performance before and after passivation, relative Co₂C/Co⁰ ratio calculated by XRD and LCF of XANES, ASF distributions, 120 h of time on stream of Co1Mn/AC catalyst, H₂-TPR profiles, HRTEM images, XRD patterns, STEM-EDS elemental maps, in situ XRD pattern of calcined Co1Mn/AC catalyst in H₂ and syngas atmosphere, oxidation state of Co species in spent Co1Mn/AC catalyst, data fitting of k²-weighted EXAFS Fourier transform magnitude spectra of Co1Mn/AC-130 sample, Mn K-edge XANES and its LCF of Co1Mn/AC-Red sample, C 1s and O 1s XPS spectra of Co/AC and Co1Mn/AC samples (PDF)

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Notes

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■ REFERENCES

- (1) Dry, M. E. *Catal. Today* **2002**, 71, 227–241.
- (2) Zhong, L.; Yu, F.; An, Y.; Zhao, Y.; Sun, Y.; Li, Z.; Lin, T.; Lin, Y.; Qi, X.; Dai, Y.; Gu, L.; Hu, J.; Jin, S.; Shen, Q.; Wang, H. *Nature* **2016**, 538, 84–87.
- (3) Fang, K.; Li, D.; Lin, M.; Xiang, M.; Wei, W.; Sun, Y. *Catal. Today* **2009**, 147, 133–138.
- (4) Xiang, Y.; Kruse, N. *Nat. Commun.* **2016**, 7, 13058.
- (5) Luk, H. T.; Mondelli, C.; Ferre, D. C.; Stewart, J. A.; Perez-Ramirez, J. *Chem. Soc. Rev.* **2017**, 46, 1358–1426.
- (6) Iglesia, E. *Appl. Catal., A* **1997**, 161, 59–78.
- (7) Khodakov, A. Y.; Chu, W.; Fongarland, P. *Chem. Rev.* **2007**, 107, 1692–1744.
- (8) Fischer, N.; van Steen, E.; Claeys, M. J. *Catal.* **2013**, 299, 67–80.
- (9) Bezemer, G. L.; Bitter, J. H.; Kuipers, H. P.; Oosterbeek, H.; Holeywijn, J. E.; Xu, X.; Kapteijn, F.; van Dillen, A. J.; de Jong, K. P. *J. Am. Chem. Soc.* **2006**, 128, 3956–3964.
- (10) Volkova, G. G.; Yurieva, T. M.; Plyasova, L. M.; Naumova, M. I.; Zaikovskii, V. I. *J. Mol. Catal. A: Chem.* **2000**, 158, 389–393.
- (11) Xiang, Y.; Chitry, V.; Liddicoat, P.; Felfer, P.; Cairney, J.; Ringer, S.; Kruse, N. *J. Am. Chem. Soc.* **2013**, 135, 7114–7117.
- (12) Anton, J.; Nebel, J.; Song, H.; Froese, C.; Weide, P.; Ruland, H.; Muhler, M.; Kaluza, S. J. *Catal.* **2016**, 335, 175–186.
- (13) Wang, Z.; Kumar, N.; Spivey, J. J. *J. Catal.* **2016**, 339, 1–8.
- (14) Pei, Y.; Liu, J.; Zhao, Y.; Ding, Y.; Liu, T.; Dong, W.; Zhu, H.; Su, H.; Yan, L.; Li, J.; Li, W. *ACS Catal.* **2015**, 5, 3620–3624.
- (15) Claeys, M.; Dry, M. E.; van Steen, E.; du Plessis, E.; van Berge, P. J.; Saib, A. M.; Moodley, D. J. *J. Catal.* **2014**, 318, 193–202.
- (16) Du, H.; Zhu, H.; Chen, X.; Dong, W.; Lu, W.; Luo, W.; Jiang, M.; Liu, T.; Ding, Y. *Fuel* **2016**, 182, 42–49.
- (17) Jiao, G.; Ding, Y.; Zhu, H.; Li, X.; Li, J.; Lin, R.; Dong, W.; Gong, L.; Pei, Y.; Lu, Y. *Appl. Catal., A* **2009**, 364, 137–142.
- (18) Lebarbier, V. M.; Mei, D.; Kim, D. H.; Andersen, A.; Male, J. L.; Holladay, J. E.; Rousseau, R.; Wang, Y. *J. Phys. Chem. C* **2011**, 115, 17440–17451.
- (19) Weller, S.; Hofer, L. J. E.; Anderson, R. B. *J. Am. Chem. Soc.* **1948**, 70, 799–801.
- (20) Xiong, J.; Ding, Y.; Wang, T.; Yan, L.; Chen, W.; Zhu, H.; Lu, Y. *Catal. Lett.* **2005**, 102, 265–269.
- (21) Li, Z.; Zhong, L.; Yu, F.; An, Y.; Dai, Y.; Yang, Y.; Lin, T.; Li, S.; Wang, H.; Gao, P.; Sun, Y.; He, M. *ACS Catal.* **2017**, 7, 3622–3631.
- (22) Weisz, P. B.; Prater, C. D. *Adv. Catal.* **1954**, 6, 143–196.
- (23) Du, H.; Zhu, H.; Liu, T.; Zhao, Z.; Chen, X.; Dong, W.; Lu, W.; Luo, W.; Ding, Y. *Catal. Today* **2017**, 281, 549–558.
- (24) Dong, W.; Liu, J.; Zhu, H.; Ding, Y.; Pei, Y.; Liu, J.; Du, H.; Jiang, M.; Liu, T.; Su, H.; Li, W. *J. Phys. Chem. C* **2014**, 118 (33), 19114–19122.
- (25) Mohandas, J. C.; Gnanamani, M. K.; Jacobs, G.; Ma, W.; Ji, Y.; Khalid, S.; Davis, B. H. *ACS Catal.* **2011**, 1, 1581–1588.
- (26) Xiang, Y.; Barbosa, R.; Kruse, N. *ACS Catal.* **2014**, 4, 2792–2800.
- (27) Boudeville, Y.; Figueras, F.; Forissier, M.; Portefaix, J. L.; Vedrine, J. C. *J. Catal.* **1979**, 58, 52–60.
- (28) Scofield, J. H. *J. Electron Spectrosc. Relat. Phenom.* **1976**, 8, 129–137.
- (29) Graham, U. M.; Jacobs, G.; Gnanamani, M. K.; Lipka, S. M.; Shafer, W. D.; Swartz, C. R.; Jermwongratnachai, T.; Chen, R.; Rogers, F.; Davis, B. H. *ACS Catal.* **2014**, 4, 1662–1672.
- (30) Nakamura, J.; Toyoshima, I.; Tanaka, K. *Surf. Sci.* **1988**, 201, 185–194.
- (31) Johnson, G. R.; Werner, S.; Bell, A. T. *ACS Catal.* **2015**, 5, 5888–5903.
- (32) Johnson, G. R.; Bell, A. T. *J. Catal.* **2016**, 338, 250–264.

- (33) Lukashuk, L.; Föttinger, K.; Kolar, E.; Rameshan, C.; Teschner, D.; Hävecker, M.; Knop Gericke, A.; Yigit, N.; Li, H.; McDermott, E.; Stöger Pollach, M.; Rupprechter, G. *J. Catal.* **2016**, *344*, 1–15.
- (34) Bezemer, G. L.; Radstake, P.; Falke, U.; Oosterbeek, H.; Kuipers, H.; Van Dillen, A.; De Jong, K. *J. Catal.* **2006**, *237*, 152–161.
- (35) Morales, F.; Degroot, F.; Gijzeman, O.; Mens, A.; Stephan, O.; Weckhuysen, B. *J. Catal.* **2005**, *230*, 301–308.
- (36) Kwak, G.; Kim, D. E.; Kim, Y. T.; Park, H. G.; Kang, S. C.; Ha, K. S.; Jun, K. W.; Lee, Y. J. *Catal. Sci. Technol.* **2016**, *6*, 4594–4600.
- (37) den Breejen, J. P.; van Schooneveld, M. M.; Stephan, O.; Frey, A. M.; Yang, J.; de Groot, F. M. F.; Bitter, J. H.; Holmen, A.; de Jong, K. P. *Top. Catal.* **2011**, *54*, 768–777.
- (38) Das, D.; Ravichandran, G.; Chakrabarty, D. K. *Catal. Today* **1997**, *36*, 285–293.
- (39) Zhu, Y.; Pan, X.; Jiao, F.; Li, J.; Yang, J.; Ding, M.; Han, Y.; Liu, Z.; Bao, X. *ACS Catal.* **2017**, *7*, 2800–2804.
- (40) Li, B.; Zhang, Q.; Chen, L.; Cui, P.; Pan, X. *Phys. Chem. Chem. Phys.* **2010**, *12*, 7848–7855.
- (41) Claeys, M. *Nature* **2016**, *538*, 44–45.
- (42) Iglesia, E.; Reyes, S. C.; Madon, R. J.; Soled, S. L. *Adv. Catal.* **1993**, *39*, 221–302.
- (43) Bezemer, G. L.; Falke, U.; van Dillen, A. J.; de Jong, K. P. *Chem. Commun.* **2005**, *6*, 731–733.
- (44) Dinse, A.; Aigner, M.; Ulbrich, M.; Johnson, G. R.; Bell, A. T. *J. Catal.* **2012**, *288*, 104–114.